Dynamic Cart Lab

Course: SPH4UE

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Image of Design



Coefficient of Friction

Collected Data

Quantity	Symbol	Value
Mass of standard cart (no crumple zone)	m_s	0.5085 kg
Ramp height	h	10.65 cm
Track length (approximate)	L	100 cm
Ramp angle	θ	6.1°
Gravitational field strength	g	9.81 m/s^2

The ramp angle comes from the height and track length:

$$\sin\theta = \frac{h}{L} = \frac{10.65}{100} = 0.1065, \qquad \theta = \sin^{-1}(0.1065) = 6.11^\circ \approx 6.1^\circ.$$

The track length was only measured to between 99 and 101 cm, so the angle is good to about $\pm 0.1^\circ$.

Table 1: Standard cart descent trials (released from rest, no crumple zone).

Trial	Time t (s)	Displacement d (m)
1	1.75	0.887
2	1.85	0.879
3	2.00	0.885
Mean	1.87	0.884

$$\bar{t}_s = \frac{1.75 + 1.85 + 2.00}{3} = 1.8667 \text{ s}, \qquad \bar{d}_s = \frac{0.887 + 0.879 + 0.885}{3} = 0.8837 \text{ m}.$$

Free-Body Diagram of the Cart

Coordinate system: $+x$ is down the ramp (parallel to the incline) and $+y$ is perpendicular to the ramp, pointing away from the surface.

Calculation for Coefficient of Friction

Newton's second law along the ramp (x-direction).

$$F_{\text{net},x} = F_{g,x} - F_k = ma_x, \qquad F_{g,x} = mg\sin\theta, \qquad F_k = \mu_k F_N.$$

Newton's second law perpendicular to the ramp (y-direction). There is no acceleration in the y-direction, so

$$F_N - mg\cos\theta = 0 \quad \Rightarrow \quad F_N = mg\cos\theta.$$

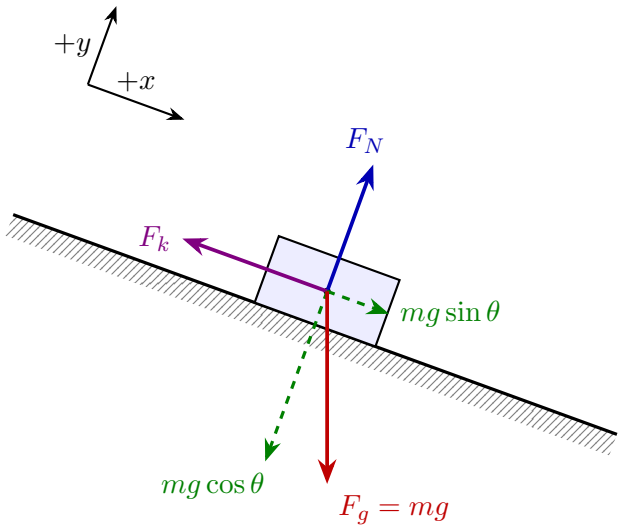


Figure 1: Forces on the cart as it rolls down the ramp. The ramp is drawn with an exaggerated angle and is not to scale.

Force components on the standard cart.

$$F_{g,x} = m_s g \sin\theta = (0.5085)(9.81)\sin 6.1^\circ = 0.530 \text{ N},$$
$$F_N = m_s g \cos\theta = (0.5085)(9.81)\cos 6.1^\circ = 4.96 \text{ N}.$$

Theoretical acceleration (no friction). The mass divides out of $mg\sin\theta = ma_{\text{theo}}$:

$$a_{\text{theo}} = g\sin\theta = 9.81\sin 6.1^\circ = 1.04 \text{ m/s}^2.$$

Experimental acceleration (from rest). Using $d = v_i t + \frac{1}{2}at^2$ with $v_i = 0$:

$$a_{\text{exp}} = \frac{2\bar{d}_s}{\bar{t}_s^2} = \frac{2(0.8837)}{(1.8667)^2} = 0.507 \text{ m/s}^2.$$

Acceleration and force due to friction.

$$a_{\text{friction}} = a_{\text{theo}} - a_{\text{exp}} = 1.0425 - 0.5072 = 0.535 \text{ m/s}^2,$$
$$F_k = m_s\, a_{\text{friction}} = (0.5085)(0.5352) = 0.272 \text{ N (up the ramp)}.$$

Coefficient of friction. Substituting $F_k = \mu_k mg\cos\theta$ and dividing by m:

$$g\sin\theta - \mu_k g\cos\theta = a_{\text{exp}} \quad \Rightarrow \quad \mu_k = \frac{g\sin\theta - a_{\text{exp}}}{g\cos\theta} = \frac{9.81\sin 6.1^\circ - 0.5072}{9.81\cos 6.1^\circ} = 0.0549.$$

Table 2: Friction results (standard cart).

Quantity	Symbol	Value
Theoretical acceleration	a_{theo}	$1.04\,\text{m/s}^2$
Experimental acceleration	a_{exp}	$0.507\,\text{m/s}^2$
Acceleration due to friction	a_{friction}	$0.535\,\text{m/s}^2$
Kinetic friction force	F_k	$0.272\,\text{N}$
Coefficient of friction	μ_k	0.0549

Impact Velocity

Collected Data

The final cart (all modifications and passengers included) has a mass of $m_f = 0.5789\,\text{kg}$. In each test the cart was released from rest at x_i and struck the barrier at $x_f = 0.995\,\text{m}$.

Table 3: Final cart descent trials.

Test	t_i (s)	t_f (s)	x_i (m)	x_f (m)	Δt (s)	Δx (m)
1	3.04	4.70	0.222	0.995	1.66	0.773
2	1.03	2.64	0.220	0.995	1.61	0.775
3	2.67	4.19	0.218	0.995	1.52	0.777

Calculation of Impact Velocity

Each cart starts from rest and accelerates at an approximately constant rate down the ramp. This acceleration already includes friction, so for each test

$$\Delta x = \tfrac{1}{2} a\,\Delta t^2 \quad \Rightarrow \quad a = \frac{2\Delta x}{\Delta t^2}, \qquad v_{\text{impact}} = a\,\Delta t = \frac{2\Delta x}{\Delta t}.$$

Worked example (Test 1):

$$a_1 = \frac{2(0.773)}{(1.66)^2} = 0.561\ \text{m/s}^2, \qquad v_{\text{impact},1} = \frac{2(0.773)}{1.66} = 0.931\ \text{m/s}.$$

Table 4: Acceleration and impact velocity of the final cart.

Test	Acceleration a (m/s^2)	Impact velocity v_{impact} (m/s)
1	0.561	0.931
2	0.598	0.963
3	0.673	1.02

Experimental Impact Force

Collected Data

Table 5: Crumple zone position readings.

Test	Position before crumple (m)	Position at max crumple (m)	Time of max crumple (s)
1	0.645	0.623	4.74
2	0.615	0.643	2.73
3	0.649	0.626	4.22*

*The Test 3 time was first recorded as 2.70 s because the crumple duration (0.03 s) was added to the release time instead of the impact time. The corrected value is $4.19 + 0.03 = 4.22$ s.

Free-Body Diagram of the Cart During Impact

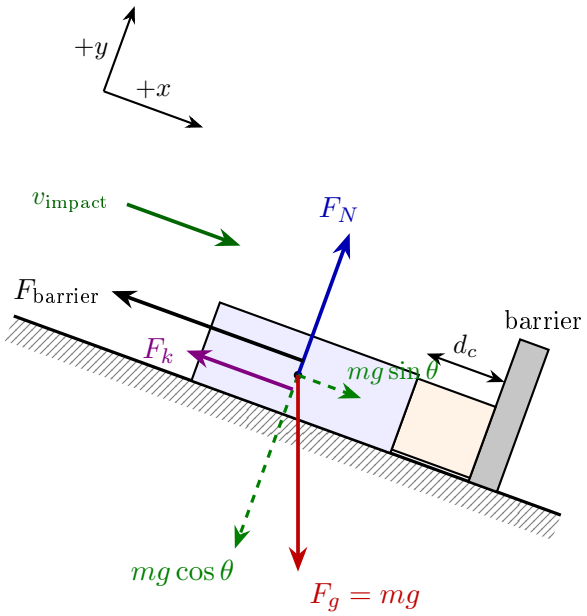


Figure 2: Forces on the cart while the barrier stops it. The crumple zone (hatched) compresses by d_c. The ramp angle is exaggerated and the diagram is not to scale.

Deformation Distance

The deformation distance is the size of the change in position:

$$d_{c,1} = |0.645 - 0.623| = 0.022 \text{ m} = 2.2 \text{ cm},$$
$$d_{c,2} = |0.615 - 0.643| = 0.028 \text{ m} = 2.8 \text{ cm},$$
$$d_{c,3} = |0.649 - 0.626| = 0.023 \text{ m} = 2.3 \text{ cm}.$$

Acceleration During Impact

Taking $+x$ down the ramp, the cart stops ($v_f = 0$) over the distance d_c, so the acceleration points up the ramp and is negative:

$$v_f^2 = v_{\text{impact}}^2 + 2a_{\text{impact}}d_c \quad \Rightarrow \quad a_{\text{impact}} = -\frac{v_{\text{impact}}^2}{2d_c}.$$

Worked example (Test 1):

$$a_{\text{impact},1} = -\frac{(0.9313)^2}{2(0.022)} = -19.7 \text{ m/s}^2 \approx -20 \text{ m/s}^2.$$

Impact Force

Along the ramp, the forces on the cart are the down-ramp component of gravity, the up-ramp friction force and the up-ramp force from the barrier:

$$F_{\text{net},x} = mg\sin\theta - F_k - F_{\text{barrier}} = ma_{\text{impact}}.$$

Before the collision the same two forces gave the cart's measured acceleration, $mg\sin\theta - F_k = ma$ (Table 4). Substituting,

$$ma - F_{\text{barrier}} = ma_{\text{impact}} \quad \Rightarrow \quad F_{\text{barrier}} = m_f\left(a - a_{\text{impact}}\right).$$

Worked example (Test 1):

$$F_{\text{barrier},1} = 0.5789\left[0.561 - (-19.713)\right] = 11.7 \text{ N} \approx 12 \text{ N (up the ramp)}.$$

This assumes that the rolling resistance acting just before the collision also acts during it.

Table 6: Impact results for the final cart.

Test	d_c (cm)	v_{impact} (m/s)	a_{impact} (m/s^2)	F_{barrier} (N)	Stopping time $2d_c/v$ (s)
1	2.2	0.931	−20	12	0.047
2	2.8	0.963	−17	9.9	0.058
3	2.3	1.02	−23	14	0.045

As a rough check, the stopping times calculated from the crumple distances (0.045 to 0.058 s) are the same order of magnitude as the durations read from the recording (0.04, 0.09 and 0.03 s). The differences, largest in Test 2, likely come from timing uncertainty or from the stopping acceleration not being constant.

Comparison to a Rigid Car

A rigid car stops over only $d_{\text{rigid}} = 0.01\,\text{cm} = 0.0001\,\text{m}$. To isolate the effect of the crumple zone, each rigid-car case uses the same mass and the same impact velocity as the matching test, so only the stopping distance changes.

Acceleration and Force of a Rigid Car

$$a_{\text{rigid}} = -\frac{v_{\text{impact}}^2}{2d_{\text{rigid}}}, \qquad F_{\text{rigid}} = m_f\,(a - a_{\text{rigid}})\,.$$

Worked example (Test 1):

$$a_{\text{rigid},1} = -\frac{(0.9313)^2}{2(0.0001)} = -4337 \text{ m/s}^2 \approx -4.3 \times 10^3 \text{ m/s}^2,$$

$$F_{\text{rigid},1} = 0.5789\,[\,0.561 - (-4336.8)\,] = 2511 \text{ N} \approx 2.5 \times 10^3 \text{ N}.$$

Percentage Reduction in Impact Force

$$\%\text{ reduction} = \frac{F_{\text{rigid}} - F_{\text{barrier}}}{F_{\text{rigid}}} \times 100\%.$$

For Test 1:

$$\frac{2510.9 - 11.7}{2510.9} \times 100\% = 99.5\%.$$

Table 7: Crumple zone compared with a rigid car.

Test	Crumple force (N)	Rigid force (N)	Reduction (%)
1	12	2.5×10^3	99.5
2	9.9	2.7×10^3	99.6
3	14	3.0×10^3	99.6
Mean			99.6

Effectiveness of Our Crumple Zone

The crumple zone compressed 2.2 to 2.8 cm, while a rigid car would stop in 0.01 cm. At the same mass and impact speed, this stretches the stopping time from about 0.0002 s to about 0.05 s and cuts the deceleration by a factor of more than 200. Since $F = ma$, the barrier force fell from about 2.5 to 3.0 kN to between 9.9 and 14 N, a mean reduction of 99.6%. The crumple zone sacrifices the front of the cart to lengthen the stop, so passengers feel a much gentler deceleration.

Additional Marks

Drawing of Design

Insert labelled drawing of the cart here

crumple zone and materials, passenger restraints, key dimensions

Compliant Design That Works

- The cart travelled 77.3 to 77.7 cm down the ramp before impact, more than the required 70 cm.
- The crumple zone compressed by 2.2 to 2.8 cm and absorbed the collision.

Intermediate values were carried at full precision. Final answers are rounded to the significant digits of the measurements (two for the crumple distances and forces, three for the rest).